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Definitions

BFS (Builders FirstSource) – BLDR.com, NYSE BLDR, the building-products, component and distribution company.

Introduction & Purpose of Case Studies

Builders FirstSource (BFS) is the largest U.S. supplier of structural building products, prefabricated components and value-added services to professional builders ¹. The company's scale and innovation—in manufactured components, digital tools and logistics—have enabled it to serve diverse customers from national production builders to custom homebuilders. The purpose of this article is to showcase real projects where BFS's capabilities solved customers' challenges. Through case studies we examine the **problem**, the **design/solution**, the **delivery process** and the **outcomes** (chain-of-thought), categorize them by project type and solution (tree-of-thought) and look ahead using SCAMPER creative thinking. These examples can inspire future customers and highlight lessons for BFS associates.

Selection Criteria for Projects

Projects were selected based on the following criteria:

1. **Scale & Impact** – Large production builders, multi-state programs or projects with measurable operational improvements (e.g., millions in savings).
2. **Complexity & Innovation** – Cases involving prefabrication, digital twins or complex logistics.
3. **Diversity** – Representation of different customer segments (production, custom, high-end) and solution categories (components, digital tools, logistics).
4. **Availability of verifiable public information** – Only projects reported in BFS publications, corporate presentations or reputable third-party sources are included and cited.

Project Profiles

Project 1 – Ready-Frame® vs. stick-built homes (Aurora, CO)

Type & Category: Single-family homes (production); solution category – prefabricated components.

Problem (Chain-of-Thought)

- Homebuilders face labor shortages and safety risks when stick-framing houses. Stick-built processes require more cutting, result in more waste and longer cycle times.

- KB Home and BFS sought objective data on whether Ready-Frame® pre-cut packages could reduce cycle time, labor and waste.

Design & BFS Role

- BFS's Ready-Frame® system ships bundles of pre-cut and labeled lumber that are **SmartBundled®** and ready for framers to assemble ². It uses **building-information modeling (BIM)** to calculate framing needs and pre-cuts parts to 1/16 inch accuracy ³.
- In 2019 BFS commissioned Construction Instruction to conduct an independent study in Aurora, Colorado. Twelve single-family homes (avg. ~2,397 ft²) were built—**six with Ready-Frame® and six using traditional stick framing** ⁴. Construction Instruction observed and timed the crews ⁵.

Delivery

- Ready-Frame® packages reduced on-site cutting and eliminated the need for extensive measuring. Bundles arrived clearly labeled, allowing framers to assemble walls and roofs quickly.
- Stick-framed houses were built side-by-side to provide a true comparison.

Outcomes & Lessons

- **Productivity:** Ready-Frame® crews framed **39 % more square footage per hour** compared with stick-framing ⁶. Homes framed with Ready-Frame® were completed **20 % faster** ⁷.
- **Efficiency:** Stick-built homes required **25 % more time for order processing and scheduling** ⁸ and one additional delivery ⁹.
- **Waste reduction:** Ready-Frame® homes used only **one dumpster haul**, whereas stick-built jobs used **one to three or more** ¹⁰. The third-party study noted that the lumber reduction equates to about **7.8 trees saved per average house** ¹¹.
- **Safety & labor:** Pre-cut packages cut ladder time by **27 %** and saw-time by **60 %** ¹². Crews used less experienced labor because the packages were simpler to assemble ¹³.
- **Lessons learned:** Prefabrication reduces cycle time, labor requirements and waste. Accurate take-offs reduce errors and rework. Ready-Frame® can address skilled-labor shortages while delivering safer jobsites.

SCAMPER ideas

- **Substitute/Modify:** Integrate **sensors** into Ready-Frame® packages to track moisture or movement, enabling quality assurance during construction.
- **Combine:** Pair Ready-Frame® with BFS digital twins to simulate assembly virtually before delivery.
- **Adapt:** Use Ready-Frame® in multi-family or light-commercial projects, adjusting bundle sizes to suit taller buildings.
- **Eliminate:** Remove manual estimating entirely by connecting home-plan submission to automatic Ready-Frame® take-off calculations.

Project 2 – Digital platform customer cases (Regional Production & Custom Builders)

Type & Category: Multiple segments – production, custom, high-end custom (single-family and spec); solution category – digital tools and design services.

BFS launched **myBLDR.com**, a digital platform that integrates plan intake, digital twin generation, Home Configure, product catalog and material quoting ¹⁴ . During the 2025 corporate presentation the company highlighted several customer case studies illustrating how digital tools address distinct challenges ¹⁵ .

Builder segment	Price range & starts (per year)	Challenges	BFS digital-tool solution	Outcome
Regional production builder (GA/NC/SC/VA)	\\$400–900 K per home; ~1,200 starts	Needed to differentiate brand, increase sales, improve buyer transparency and streamline design ¹⁶	Leveraged digital tools to streamline product selection and ensure consistency, enabling shorter design cycles , reduced rework and higher customer engagement ¹⁷	Higher sales conversion; customers experienced smoother design process and fewer changes (qualitative).
Regional custom builder (MI & WI)	\\$350 K–1 M; ~200 starts	Wanted to attract high-end customers with personalized options and allow buyers to view selections online ¹⁸	Created a digital design studio using BFS tools. Photorealistic 3-D visualizations served as a virtual sales and decision-support center ¹⁹	Improved buyer conversion rates; customers could visualize and change finishes before construction.
National spec builder (operates nationally)	\\$200–600 K; ~6,500 starts	Sought to reduce time and rework from mistakes and complex permitting processes and ensure design changes met code ²⁰	Used digital tools to verify code compliance and improve operational efficiency and customer satisfaction ²⁰	Reduced permitting delays; fewer field corrections.
High-end custom builder (GA & SC)	\\$1–5 M; ~40 starts	Reliance on 2-D plans hindered buyer trust and accurate bidding ²¹	Digital tools resolved construction conflicts virtually and turned 2-D plans into a collaborative 3-D experience ²¹	Faster bid cycles; fewer change orders; improved homeowner engagement.

Key digital capabilities

- **3-D digital twin:** BFS tools transform 2-D construction documents into 3-D models with **360° photorealistic interior and exterior visualization** ²² . Customers can configure materials, finishes and structural options in real time ²² . Renderings, videos and virtual tours generate excitement and help sell homes ²³ .

- **Virtual conflict resolution:** The Build Optimize service uses the digital twin to resolve clashes among foundations, framing, HVAC, plumbing, electrical and fire systems before construction ²⁴ . Collision detection reduces rework.
- **Shoppable digital twin:** The platform connects homebuyer selections to required material lists so that each choice corresponds to a **shippable SKU** ²⁵ . Builders get accurate quotes and can integrate windows and doors automatically ²⁶ .
- **Project management:** myBLDR.com offers a comprehensive project-management platform that allows collaboration from plan intake through material procurement and jobsite management ²⁷ .

Lessons & Best Practices

- Digital tools should be **tailored** to builder segment. Production builders benefit from streamlined product selection; custom builders need photorealistic options; spec builders require compliance and efficiency; high-end builders demand immersive 3-D experiences.
- Early integration of design with procurement ensures that changes are reflected in material lists, avoiding cost overruns.
- Educating the builder's team on digital adoption is critical; BFS's case studies show that buyer engagement increases when homeowners can "see" their home virtually ²¹ .

SCAMPER ideas

- **Combine:** Integrate **Ready-Frame®** data directly into digital twins so the framing package is automatically optimized when the plan changes.
- **Adapt:** Apply digital tools to remodels or repairs, allowing homeowners to visualize additions or energy-efficiency upgrades.
- **Modify:** Expand the "Shoppable Digital Twin" to include **IoT-enabled products** (smart HVAC, sensors) so selections influence long-term energy use.

Project 3 – Fleet graphics rebrand after BFS–BMC merger

Type & Category: Logistics/branding project; solution category – corporate marketing and fleet management.

Problem

When Builders FirstSource merged with Building Materials Holding Co. (BMC) in January 2021, it became the largest supplier of structural building products ²⁸ . The combined company operated a transitory fleet of **2,813 vehicles across 16 states**—including tractors, pickups, flatbeds, box trucks and service vans. A new brand identity meant every vehicle needed updated graphics. BFS wanted minimal downtime because the fleet was constantly on the move ²⁹ .

Design & BFS Role

- BFS selected Signature Graphics as its partner due to their ability to handle large-scale national projects ³⁰ . The team recommended a mix of **spot graphics and partial wraps**, with full custom wraps for specific vehicles ³¹ .

- Bold, eye-catching designs with minimal details ensured the brand would be visible even at highway speeds ³² . Signature Graphics adhered to BFS's brand guidelines and used 3M materials for longevity ³³ .
- Project management emphasized communication: daily updates kept BFS informed, and scheduling minimized downtime ³⁴ .

Delivery & Outcome

- The entire fleet was **rebranded within the established timeline** and with minimal operational disruption ³⁵ . BFS's marketing manager praised the partner, noting that Signature Graphics "was an extension of our marketing team" ³⁶ .

Lessons & Best Practices

- Large-scale rebranding requires meticulous scheduling and communication. Partnering with a specialist ensures design consistency across diverse vehicle types.
- Using partial wraps and spot graphics can control costs while maintaining brand visibility.

SCAMPER ideas

- **Combine:** Integrate **QR codes** or NFC tags into fleet graphics so customers can scan and access digital catalogs or myBLDR.com.
- **Adapt:** Apply similar branding strategies to jobsite signage and customer communications for consistent branding.

Project 4 – Operational footprint optimization post-merger (Atlanta area)

Type & Category: Operational / logistics; solution category – consolidation and automation.

Problem

After the BFS–BMC merger, the combined entity in Atlanta operated **12 distribution yards, 4 truss plants and 4 millwork plants** ³⁷ . The overlap increased overhead and complexity. BFS wanted to reduce costs while maintaining growth capacity.

Design

- BFS executed a consolidation program: **closed three overlapping distribution yards and two component plants** while investing in automation to retain and expand capacity ³⁸ .
- The company built a **new state-of-the-art millwork plant**, replacing a smaller site. Advantages included an optimized layout and shifting lumber re-loading from third-party vendors to internal control ³⁹ .

Delivery & Outcome

- Annual overhead cost reduction: **\\$2.5–3.0 million** ⁴⁰ .
- Truss direct labor efficiency improved **>15 %** ⁴¹ .
- Working-capital days reduced by **15 %** ⁴² .

Lessons & Best Practices

- Consolidation following a merger can yield substantial savings without impairing capacity if investments in automation accompany closures.
- Centralizing millwork operations reduces third-party reliance and improves process control.

SCAMPER ideas

- **Rearrange:** Implement predictive analytics to schedule production across fewer plants, reducing idle time.
- **Modify:** Expand automation to wall panel and truss fabrication; integrate robots for improved safety.
- **Eliminate:** Remove paper-based scheduling by integrating plant operations into myBLDR.com so customers see real-time manufacturing slots.

Exemplar Project Data Table

Project name	Location	Size / units	Key innovation	Outcome metric
Ready-Frame® case study (KB Home)	Aurora, CO	12 single-family homes (avg. 2,397 ft ²) ⁴³	Pre-cut framing packages using BIM	39 % more sq-ft framed per hour; 20 % faster completion; 60 % reduction in saw-time ⁴⁴
Digital platform customer cases	GA/NC/SC/VA; MI/WI; national; GA/SC	Production builder: 1,200 starts; custom builder: 200 starts; spec builder: 6,500 starts; high-end builder: 40 starts ¹⁵	myBLDR.com with digital twins, Home Configure and shoppable material lists	Shorter design cycles, reduced rework and higher buyer engagement ⁴⁵
Fleet graphics rebrand	16 U.S. states	2,813 vehicles (mixed fleet) ⁴⁶	Spot graphics, partial and full wraps; high-quality 3M materials; coordinated scheduling	Full fleet rebranded on schedule with minimal downtime ³⁵
Operational footprint optimization	Atlanta, GA region	Consolidation of 12 yards, 4 truss plants & 4 millwork plants ⁴⁷	Consolidated facilities; new automated millwork plant	\\$2.5–3.0 M annual overhead reduction; >15 % labor efficiency improvement ⁴⁸

Lessons Learned & Best Practices

- **Prefabrication and digital tools complement each other.** Ready-Frame® demonstrates that pre-cut framing significantly reduces cycle time, waste and injuries ⁴⁴. When combined with digital design (Project 2), builders can optimize framing packages before materials are ordered, further decreasing rework.
- **Customer experience matters.** Photorealistic 3-D visualization and the ability to select materials online build buyer confidence and reduce mid-project changes ²². Builders who adopted BFS's digital tools report higher engagement and conversion rates ⁴⁵.
- **Strong communication and scheduling are critical for large programs.** The fleet rebrand succeeded because of daily updates and clear scheduling ³⁴. Similar discipline is needed when coordinating digital tool adoption across multiple stakeholders.
- **Invest in operations after mergers.** Consolidation and automation can yield significant savings without sacrificing capacity ⁴⁸. The key is to optimize the footprint while investing in technology that scales.

How These Projects Demonstrate BFS Capabilities

1. **Manufactured Components & Building Science** – Ready-Frame® case demonstrates BFS's leadership in prefabricated framing, reducing labor and waste while enhancing safety. The CSR report confirms that Ready-Frame® enables up to **20 % faster construction** and a **39 % productivity increase** per labor hour ⁴⁹.
2. **Digital Innovation** – myBLDR.com and associated digital tools showcase BFS's investment in digital transformation. The tools convert 2-D plans into 3-D digital twins with photorealistic visualization ²², allow customers to configure materials, and tie selections directly to material lists ²⁵.
3. **Operational Excellence & Logistics** – The Atlanta consolidation case underscores BFS's ability to optimize its footprint post-merger, generating millions in savings and improving labor efficiency ⁴⁸. The fleet rebranding project shows BFS's capability to manage large-scale logistics and marketing programs across multiple states ⁴⁶.
4. **Sustainability & ESG** – Ready-Frame® reduces lumber usage by about **7.8 trees per home** ¹¹ and decreases waste hauls. BFS's CSR report notes that Ready-Frame® and other manufactured components saved over **1.5 million trees** through reduced lumber consumption ⁵⁰.

Future Inspiration & SCAMPER Opportunities

The case studies above illustrate BFS's progress but also inspire future innovation:

- **Substitute / Modify** – Integrate **smart sensors** into Ready-Frame® bundles to monitor moisture or structural stress during construction. Modify digital tools to include energy-efficiency simulations and carbon-footprint calculators.
- **Combine** – Merge digital twins with IoT devices installed in prefabricated components. A combined system could track real-time performance and inform maintenance schedules.
- **Adapt** – Apply digital platforms and prefabrication to **multi-family** or **light-commercial** projects, adjusting bundle sizes and digital catalog options to support larger buildings. Adapt digital twin tools to renovation projects so homeowners can visualize improvements.
- **Put to Another Use** – Use the **shoppable digital twin** for after-sale services (e.g., remodels, warranty claims) by providing an exact virtual model of the built home.

- **Eliminate** – Remove paper-based estimating and manual data entry by integrating myBLDR.com directly into builders' ERP systems. Use AI to suggest optimal material substitutions when supply chain disruptions occur.
- **Rearrange** – Sequence construction tasks based on digital-twin analysis to improve jobsite logistics. Rearrange the supply chain to deliver components "just-in-time" using BFS's logistics network.

Likely User Questions

1. **What was the largest multi-family or production job BFS has supplied with Ready-Frame®?** – BFS primarily cites single-family studies; further research from BFS digital tools or Ready-Frame® team may reveal multi-family applications.
2. **How much time is saved by using Ready-Frame® compared with stick framing?** – Independent studies show Ready-Frame® projects can reduce cycle time by **20 %** and enable **39 % more productivity per labor hour** ⁵¹ .
3. **What cost savings are realized by digital tools?** – Builders report shorter design cycles and fewer change orders ⁴⁵ . Savings come from reduced rework, faster permitting and accurate material lists; exact amounts depend on project size.
4. **How does the shoppable digital twin work?** – The digital twin links customer selections (siding, roofing, finishes) to a **material list** and provides a shippable SKU, allowing accurate quotes and procurement ²⁵ .
5. **Can Ready-Frame® be used in custom or high-end projects?** – Yes. The KB Home study focused on production homes, but Ready-Frame® is also used by high-end custom builders; its benefits scale with project complexity.
6. **What is myBLDR.com and how is it different from other builder portals?** – myBLDR.com integrates plan submission, 3-D digital twin creation, material quoting and product catalog in one platform ²⁷ . It provides photorealistic visualization and ties selections to material lists.
7. **How do digital tools improve sustainability?** – By enabling precise take-offs and prefabrication, digital tools reduce waste and cut delivery trips. BFS estimates that Ready-Frame® and manufactured components avoided **~220,000 tons of CO₂e** in 2023 ⁵² .

Related Topics / Crosslinks

- **Digital Tools** – deeper dive into myBLDR.com, Home Configure, Build Optimize and Render.
- **Ready-Frame® & Components** – technical overview of pre-cut framing, trusses and wall panels.
- **Sustainability & CSR** – BFS's environmental initiatives, including tree-saving metrics and partnerships (e.g., Weyerhaeuser).
- **Operational Excellence** – BFS's 1-TEAM operating system, Lean practices and automation investments.
- **Merger Integration** – how the BFS-BMC merger created the largest U.S. supplier and lessons from integrating two large organizations.

¹ ¹⁵ ¹⁶ ¹⁷ ¹⁸ ¹⁹ ²⁰ ²¹ ⁴⁵ Builders FirstSource Corporate Presentation
https://s202.q4cdn.com/867695273/files/doc_presentations/2025/May/01/BLDR-Corporate-Presentation-May-2025.pdf

² ⁴ ⁵ ⁶ ⁷ ⁸ ⁹ ¹⁰ ¹² ¹³ ⁴³ ⁴⁴ ⁵¹ BLDR_Ready-Frame_Case%20Study_1
https://bmc.a.bigcontent.io/v1/static/BLDR_Ready-Frame_Case%20Study_1

3 11 **EX-99.1**

<https://www.sec.gov/Archives/edgar/data/1316835/000119312521349718/d254698dex991.htm>

14 22 23 24 25 26 27 **PowerPoint Presentation**

<https://myparadigm.com/wp-content/uploads/2023/11/Paradigm-Annual-Conference-2023-Digital-Tools-for-Home-Builders.pdf>

28 29 30 31 32 33 34 35 36 46 **Case Study: Builders FirstSource Merger & Fleet Graphics Rebrand - Signature Graphics**

<https://signaturegraphicsinc.com/case-study-builders-firstsource-merger-fleet-graphics-rebrand/>

37 38 39 40 41 42 47 48 **Builders FirstSource Investor Day 2023**

https://s202.q4cdn.com/867695273/files/doc_presentations/2023/12/BLDR-2023-Investor-Day-Presentation-Final.pdf

49 50 52 **bfs-2024-csr-report.pdf**

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